Charcoal Smoker Controller Phase 1 Design Package

Printed 2026-09-20. Project: charcoal-smoker.

1. Current System Baseline (Phase 0)
2. Phase 1 Architecture
3. Phase 1 First Purchase List
4. Phase 1 Electrical Wiring (notes)
5. Electrical Wiring Diagram (landscape)

Charcoal Smoker - Current System Baseline (Phase 0)

Captured: 2026-09-20
Status: as-built reference before Phase 1 (ESP32-S3 + PWM + BLE) redesign
Sources: vendor technical reply (기술제안서 답변, project files charcoal-smoker-v1-pt1..pt4.png), Jason's photos (2026-09-20_01_01_*.png, annotated drum photo), Gumsung Controls catalog (gscontrol.kr, fetched 2026-09-20)

0. PROTOTYPE SCOPE (decided 2026-09-20)

The prototype platform is the stainless DRUM smoker (Jason's garage unit), not the vendor's rotisserie cabinet. The drum unit has NO 220 V PSU, NO transformer / 12 V rail, NO lights, NO rotation motor. Electrical content of the prototype platform is therefore only:

- Gumsung capillary thermostat (label "03-3203 5", see section 3) in its stainless dial enclosure, clamped to the left vertical support post.
- Capillary bulb in a brass compression fitting on a stainless block, inserted at the small port on the drum wall beside the dial thermometer (center of cooking zone height).
- 6 V brushed blower (800 mA max) fed by the 6 V / 800 mA adapter (or 4 x AA in the camping variant), nozzle into the fire door / charcoal basket area.

Sections 1 and 4 below describe the vendor's cabinet product and are reference only; they do not apply to the prototype hardware.

DECISION 2026-09-20: SINGLE SUPPLY. One external DC adapter feeds the controller board; the board derives the blower rail (6 V) and the logic rail (3.3 V / 5 V) on-board. The existing 6 V / 800 mA adapter is retired from the prototype.

DECISION 2026-09-20: KEEP THE EXISTING THERMOSTAT as the Phase 1 hardware over-temp cutoff. It meets the role (mechanical, independent of software, contact rating far above the 6 V / 0.8 A blower load, 50 to 320 C range covers a cutoff above any cook target). Replace only if the capillary or bulb is damaged.

Consequences for Phase 1 design:

- No 12 V rail exists on the prototype; all rails come from the single adapter via on-board regulators.
- Mounting: the thermostat enclosure position on the post is a candidate location for the controller enclosure; the probe port is the candidate location for the digital probe.

1. Power system (vendor reply Q1, Q5) - cabinet product, reference only

- Mains input: AC 220 V / 60 Hz (rated supply).
- 220 V is used directly only for the power indicator lamp.
- All other electrics run from a transformer producing DC 12 V: internal rotation DC motor (rotisserie), internal halogen lamps, toggle switches.

- Blower: originally 4 x AA. Modified to a DC 6 V / 800 mA (rated output) wall adapter plugged into the 220 V outlet.
- Blower spec per vendor: voltage 6 V, max current 800 mA, motor type DC BRUSHED (DC브러쉬드).
- Blower does NOT run from the 12 V rail.

CORRECTION vs earlier summary: the vendor states the blower motor is brushed, not brushless. This removes the supply-side PWM concern; a plain low-side MOSFET PWM stage with flyback diode is appropriate.

2. Blower control (vendor reply Q2) - applies to prototype

- ON/OFF only. No PWM.
- Below setpoint -> blower ON. Setpoint reached -> thermostat contact opens -> blower OFF.
- The dial on the stainless box is the thermostat setpoint knob (0, 50 to 320 C), not a timer. No timer exists in the system.

3. Temperature controller (vendor reply Q3 + Gumsung catalog) - applies to prototype

- Mechanical capillary thermostat with dial knob, rated 250 V / 16 A.
- Manufacturer: Gumsung Controls Co (금성콘트롤), Korea. Catalog: gscontrol.kr.
- Label, as read from the unit: 모델명: 03-3203 5. Gumsung's public catalog does not list this string; the matching catalog part by range and rating is GS-320S (50 to 320 C, 250 VAC/16 A). Treat "03-3203 5" as the label text and GS-320S as the catalog equivalent.
- Label also shows: E239121 beside a CE mark (UL file-number format, not verified), 안전인증번호 XD020003-15001A (KC safety cert), mfg 2025.11, 정격 250V-16A, 제조국 한국.
- Catalog spec for GS-320S: range 50 to 320 C, differential 10 +/- 5 C, contact ratings 220 VAC/24 A, 250 VAC/16 A, 400 VAC/10 A, contact type SPST or SPDT, capillary 1000 mm standard (150 to 2000 mm option), bulb 5 mm dia x 65 mm or 3 mm dia x 169 mm.
- Three 6.3 mm spade terminals on the body, consistent with the SPDT variant (common + NC + NO). Which terminal is which: not verified; confirm by continuity at room temperature.
- Differential of 10 +/- 5 C means the failsafe will re-close roughly 5 to 15 C below the dial value once tripped.
- No MCU, no digital sensing, no PID, no software.

4. Sensor location (vendor reply Q4) - cabinet product

- Bulb positioned at the center of the cooking zone inside the cabinet.
- Mounting: a small stainless bracket/clip on the cabinet wall holds the bulb; the dial enclosure mounts on the vertical support post outside. Vendor states removable (탈부착 가능).
- The product is a rotating smokeless charcoal BBQ cabinet: glass door, halogen lamps top, toggle switches and pilot lamp on the top panel, side thermometer, blower + sensor on the post, valve, rotisserie racks, charcoal basket, brake casters.
- On the prototype drum the same parts mount as described in section 0.

5. Two blower configurations (vendor reply note)

- Same sensor, same ON/OFF method, same temperature setting on both.
- Camping/home: 4 x AA. Restaurant/commercial: adapter.

6. Disposition of existing controller (vendor reply Q6)

- Vendor: replacement or parallel use is possible (가능).
- Chosen direction: remove from normal regulation, keep in series in the blower power path as an independent hardware over-temperature cutoff.

7. Photo observations (2026-09-20)

- Prototype drum: stainless drum, two vertical stainless tubes from the lid (unused on this unit; no motor), Celsius dial thermometer approx 50 to 350 C, small open port beside it (probe port), hinged fire door with mesh charcoal basket, drip tray, casters.
- Smaller second unit (not the prototype): stainless drum, glass lid, small bimetal gauge, hinged door; black red-fronted box on left post, identity not confirmed.
- Thermostat enclosure: stainless box, black setpoint dial, cable gland, gray lead to blower. Clamps to the drum's left post.
- Blower: black centrifugal, straight metal outlet nozzle, mesh intake, plastic handle, DC barrel jack on handle. Vendor photos show the AA compartment in the handle rewired to the jack.
- Probe: stainless bulb in a brass compression fitting on a stainless block, capillary to thermostat. Unscrews.

Current architecture (prototype drum)

```
6 V 800 mA adapter (or 4 x AA)
   -> thermostat contact (ON/OFF, dial = setpoint)
   -> 6 V brushed blower -> charcoal basket -> chamber temp -> capillary bulb at side port
(feedback)
```

Gaps (what the current unit lacks)

No PWM, no variable airflow, no MCU, no digital temperature, no PID, no state machine, no ramp-up/stabilize modes, no lid-open detection, no history, no digital setpoint, no fault/event log, no BLE, no software safety logic.

Phase 1 target (center of architecture)

Single DC adapter -> controller board (6 V blower rail + logic rail) -> ESP32-S3 (control + state) -> low-side MOSFET PWM -> 6 V brushed blower, with the existing thermostat retained in series in the blower rail as hardware over-temp cutoff (dial set above max cook temperature). Digital temp sensor at the probe port.

Open items to verify before Phase 1 build

- Thermostat terminal pinout (which pair is closed below setpoint). Continuity check.
- Actual blower running current at 6 V (vendor gives 800 mA max only).
- Probe port: thread of the brass compression fitting and diameter of the side port.

Charcoal Smoker - Phase 1 Architecture (ESP32-S3 + PWM + BLE)

Created: 2026-09-20
Status: DRAFT. Builds on the Phase 0 baseline. Decisions marked DECIDED are fixed; OPEN items need a measurement or a choice before build.
Platform: stainless drum smoker, no PSU / lights / motor. Only loads: blower + controller.

1. Fixed decisions

- DECIDED: single external 12 V DC adapter; all rails derived on the controller board.
- DECIDED: existing Gumsung capillary thermostat (label 03-3203 5, catalog equivalent GS-320S) stays in series in the blower rail as the hardware over-temp cutoff. Dial set above max cook target. It never feeds the controller's supply.
- DECIDED: existing 6 V brushed blower (800 mA max) retained, driven by PWM.
- DECIDED: ESP32-S3 as MCU (PWM via LEDC peripheral, BLE 5 LE; no Bluetooth Classic on S3).

2. Power tree

```
12 V 2 A adapter
  |-- reverse-polarity protection + 2 A input fuse
  |-- Buck A (LM2596) -> 6.0 V blower rail
  |       -> thermostat contact (series, hardware cutoff)
  |       -> blower (+)
  |       blower (-) -> low-side N-MOSFET -> GND   [flyback diode across blower]
  |-- Buck B (LM2596) -> 5.0 V logic rail
          -> ESP32-S3 5V pin (budget 0.5 A peak during BLE TX)
          -> ESP32 3V3 -> MAX31856 sensor front end
```

Budget: blower 0.8 A at 6 V = 4.8 W; logic 0.5 A at 3.3 V = 1.7 W; regulator losses about 15 percent. A 12 V / 2 A adapter has ample margin for a display or second sensor later. Rationale for 12 V input: keeps both bucks in step-down mode with headroom, common adapter, leaves room for a 12 V accessory later.

3. Safety chain (hardware, no software in the loop)

1. Thermostat contact in series with the blower rail. Opens above dial setpoint. Re-closes 5 to 15 C below (10 +/- 5 C differential). Wire the pair that is closed at room temperature (OPEN: continuity check identifies it).
2. Input fuse sized to adapter rating.
3. MOSFET gate pull-down so the blower is OFF while the ESP32 is in reset, booting, or unpowered.
4. Optional: thermal fuse or a second fixed-temperature bimetal switch on the drum wall for unattended long cooks.

Software adds on top: max duty clamp, sensor-fault -> blower OFF, watchdog reset -> blower OFF, over-temp alarm at a lower threshold than the thermostat dial so the controller acts first and the thermostat is the backstop.

4. Blower drive

- Topology: low-side N-channel logic-level MOSFET (IRLZ44N), 200 ohm gate resistor, 10 k gate pull-down, 1N5819 Schottky flyback diode across the motor terminals (cathode to +6 V).
- PWM source: ESP32-S3 LEDC, one channel.
- Frequency: OPEN. Bench-test two candidates on the real blower: about 25 kHz (inaudible, smoother current) and about 200 Hz (better starting torque at low duty, may whine). Pick by measuring the minimum duty that reliably spins the blower from rest.
- Minimum duty: map it (OPEN) and enforce in software: duty command 0 -> OFF; 1..100 percent scaled into [min_start_duty, 100]. Consider a short 100 percent kick on start.
- Current sense (optional): low-side shunt + ADC to detect a stalled or disconnected blower.

5. Temperature sensing

- Primary: K-type thermocouple at the side port beside the dial thermometer, MAX31856 SPI front end (cold-junction compensated, 50/60 Hz filter, fault register).
- Accuracy class +/- 2 C is adequate for pit control at 90 to 150 C. An RTD (MAX31865 class) is a drop-in Phase 2 swap since both are SPI.
- Thermocouple junction must be ungrounded; the drum is metal.
- Secondary: meat probes on the second MAX31856 or the MAX31855 boards, display only, not in the control loop.
- Sampling: 1 Hz for control, median-of-5 filter, flag open/short from the fault bits.

6. Control loop and state machine

- Loop: PID on pit temperature, output = blower duty. Sample 1 s, update duty 1 s. Anti-windup on the integral. Derivative on measurement, not error.
- States: IDLE (blower off, logging), IGNITE (fixed high duty until pit crosses setpoint minus margin, or timeout -> FAULT), RAMP (PID clamped high, exits to HOLD within +/- 5 C for N seconds), HOLD (full PID), LID_OPEN (rapid drop detected: freeze integral, hold or floor the duty, exit on recovery or timeout), FAULT (blower off, alarm: sensor open/short, no rise during IGNITE, over software limit, watchdog).
- Tuning: log a cook at fixed duty steps first (step response), derive initial PID from that, then refine.

7. BLE

- ESP32-S3 as GATT server, phone as client.
- One custom service: pit temp (notify), meat temps (notify), setpoint (read/write), blower duty (read), state (notify), fault code (notify), event log (read, chunked), device info.
- Phase 1 client: web BLE page or a generic BLE scanner app; native app later.

- Security: pairing enabled for 60 s by a physical button; no setpoint write without a bonded client.

8. Firmware stack

- Framework: ESP-IDF or Arduino-ESP32 (OPEN; Arduino is faster to bring up with the Adafruit_MAX31856 library, ESP-IDF gives cleaner BLE and watchdog control).
- Modules: sensor driver, PID + state machine, blower driver, BLE service, event log (ring buffer in flash), config store (NVS), OTA (Phase 2).
- Watchdog on the control task; on reset the gate pull-down guarantees blower OFF.

9. Enclosure and mounting

- Controller in a sealed box clamped to the left post beside the thermostat box. Keep the thermostat box as-is; only its contact wiring changes.
- Cable from controller to blower: 2-wire 6 V PWM rail through the thermostat contact. Reuse the barrel jack on the blower handle.
- Probe cable entry via cable gland; thermocouple wire routed away from the blower rail.

10. Bring-up order

1. Bench: adapter + bucks, verify rails under a resistor dummy load.
2. Blower on MOSFET with bench PWM; find min start duty and choose PWM frequency.
3. Sensor front end reading in boiling water and ice water.
4. Thermostat continuity check, wire into blower rail, verify it opens with heat on the bulb.
5. Firmware: manual duty over BLE, log temperature. No PID yet.
6. Step-response cook. Derive PID. Enable HOLD.
7. Add IGNITE / LID_OPEN / FAULT logic. Full cook test.

11. Open items (blocking build)

- Thermostat terminal pair.
- Blower min start duty and PWM frequency (bench test).
- Blower running current at 6 V.
- Probe fitting thread and port diameter.
- Framework choice (ESP-IDF vs Arduino-ESP32).

Phase 1 - First Purchase List (bench bring-up)

Created: 2026-09-20. Status: ALL PARTS ORDERED.
Scope: everything needed for bring-up steps 1 to 5 (rails, blower on PWM, sensor reading, thermostat wired as cutoff, manual duty over BLE). Enclosure, PCB, and meat probes are deferred.
Already on hand: 6 V brushed blower, Gumsung capillary thermostat, drum smoker, breadboard, jumper/hookup wires, 200 ohm and 10 k resistors, multimeter, USB-C cable, heat source.

Purchase status

ORDERED, Amazon, 2026-09-19:

- YEJMKJ ESP32-S3 N8R8 2-pack (order 111-7872990-3981853, arriving Wed 2026-09-23). 1 in use, 1 spare.
- Generic MAX31856 breakout 2-pack (arriving 2026-09-29 to 10-02). 1 in use, 1 spare.
- HUABAN MAX31855 2-pack (order 111-5592486-4043419, arriving Wed 2026-09-23). Surplus to design; spare / meat-probe use.

ORDERED, Amazon, 2026-09-20:

| ASIN | Item | Price |

|---|---|---|

| B0FMYHCB6X | Bridgold IRLZ44N 10-pack | $4.49 |

| B00YX8XCZK | K-type EGT probe, 1/8 NPT compression. Junction type not stated; test on arrival (sheath-to-lead continuity = grounded) | $3.80 |

| B079KG1TN2 | Chanzon 1N5819 Schottky diodes | $7.99 |

| B0DBVYP91F | Seloky LM2596 buck 5-pack | $7.99 |

| B013HJI0Q6 | 12 V 2 A adapter, 5.5 x 2.1 mm, UL | $9.99 |

| B0BD46CP5Y | DaierTek 5.5 x 2.1 mm panel jack + plug, 6 sets | $9.99 |

| B0813Q4S6P | BOJACK inline 5 x 20 mm fuse holders + 150 fuses | $11.99 |

| B01G4POUAU | BAOMAIN 1/4 in female spade terminals, 100 pk | $6.99 |

| B0FW5C1381 | DROK K-type bench probes 3-pack | $6.99 |

Fallback pit probe if B00YX8XCZK tests grounded or reads noisy: Procon B09HSJ4PBB, $19.99, stated ungrounded, 5 x 80 mm, 1/8 NPT, 2 m braided lead.

NOT PURCHASED: reverse-polarity Schottky (SS34). The 1N5819 (1 A) is marginal for the 2 A input; use two in parallel on the bench or add an SS34 later.

A. Core electronics

| # | Item | Spec / part | Qty | Notes |

|---|---|---|---|---|

| 1 | MCU dev board | YEJMKJ ESP32-S3 N8R8, DevKitC-1 clone: WROOM-1-N8R8, dual USB-C, RGB LED, headers pre-soldered | 1 (+1 spare) | Follows the Espressif DevKitC-1 pinout. Verify module label on arrival. USB-UART driver only if the COM port does not appear. |

| 2 | Thermocouple amplifier | Generic MAX31856 breakout, Adafruit 3263 layout: VIN, 3Vo, GND, SCK, SDO, SDI, CS, FLT, DRDY, onboard 3.3 V regulator, screw terminal | 1 (+1 spare) | Confirm chip marking is MAX31856. Adafruit_MAX31856 library applies. Thermocouple must be ungrounded junction. |

| 3 | Pit probe | K-type, stainless sheath, 1/8 NPT compression fitting | 1 | Verify port thread and junction type. |

| 3a | Bench thermocouples | DROK 3-pack K-type 1 m | 1 | Boiling/ice water sanity checks only. |

| 4 | Blower switch | IRLZ44N, TO-220 | 2 of 10 | Logic level; no heatsink at 0.8 A. |

| 5 | Flyback diode | 1N5819 Schottky 1 A / 40 V | 2 of 100 | Across blower, cathode to +6 V. |

| 6 | Gate resistor | 200 ohm 1/4 W (on hand) | 2 | GPIO to gate. |

| 7 | Gate pull-down | 10 kohm 1/4 W (on hand) | 2 | Gate to GND; blower off during reset. |

| 8 | Buck, blower rail | LM2596 module set to 6.0 V | 1 of 5 | Set voltage before connecting the blower. |

| 9 | Buck, logic rail | LM2596 module set to 5.0 V | 1 of 5 | Feeds dev board 5V pin. |

| 10 | Adapter | 12 V 2 A, 5.5 x 2.1 mm center positive | 1 | |

| 11 | Barrel jack | DaierTek panel jack + plug | 1 of 6 | Plug also serves the blower lead if its jack is 5.5 x 2.1 mm. |

| 12 | Input fuse | 2 A fast-blow 5 x 20 mm in inline holder | 1 | |

| 13 | Reverse polarity | SS34 3 A Schottky or P-MOSFET | 1 | NOT BOUGHT. Bench: two 1N5819 in parallel. |

B. Bench and wiring

| # | Item | Qty | Notes |

|---|------|-----|-------|

| 14 | Breadboard, 830 pt | 1 | On hand. One is sufficient. |

| 15 | Jumper / hookup wires | 1 set | On hand. |

| 17 | 6.3 mm female spade terminals | 6 of 100 | Thermostat's three tabs. |

| 18 | DC barrel plug to bare wire | 1 | From DaierTek set (OPEN: measure blower jack). |

| 19 | USB-C data cable | 1 | On hand. |

| 20 | Multimeter | 1 | On hand. |

| 21 | Heat gun or lighter | 1 | On hand. |

C. Deferred (not needed for bring-up)

- Enclosure (IP-rated box), cable glands, panel LED, button for BLE pairing.
- Meat probes (second MAX31856 or the MAX31855 boards).

- Custom PCB or protoboard for the final build.
- Current-sense shunt for blower stall detection.
- Display.

While waiting for parts

- Measure: side port diameter, brass fitting thread OD, blower barrel jack OD/ID.
- Thermostat: continuity check across the three spade tabs at room temperature to identify the closed pair.
- Decide framework: ESP-IDF vs Arduino-ESP32.
- Install toolchain on JASON-PC and confirm the USB-UART driver situation.

Phase 1 - Electrical Wiring (bench circuit)

Created: 2026-09-20. The full diagram is on the following landscape page.

Power

- 12 V 2 A adapter -> 5.5x2.1 panel jack -> 2 A fuse -> reverse-polarity Schottky (2x 1N5819 in parallel, bench) -> +12 V node
- +12 V -> LM2596 A (set 6.0 V) -> thermostat closed pair -> blower +
- +12 V -> LM2596 B (set 5.0 V) -> ESP32-S3 5V pin
- ESP32 3V3 -> MAX31856 VIN
- Common ground: adapter -, both buck GND, ESP32 GND, MAX31856 GND, MOSFET source, 10k pull-down. Star at buck outputs.

Blower loop

- blower - -> IRLZ44N drain; IRLZ44N source -> GND
- 1N5819 across blower: anode on blower -, cathode on blower +
- GPIO4 -> 200 ohm -> gate; 10 k gate -> GND

ESP32-S3 pin map (Arduino-ESP32 DevKitC-1 defaults; confirm on silkscreen)

Pin	To	Role
5V	LM2596 B OUT+	board supply
3V3	MAX31856 VIN	sensor supply
GPIO12	MAX31856 SCK	SPI clock
GPIO11	MAX31856 SDI	MOSI
GPIO13	MAX31856 SDO	MISO
GPIO10	MAX31856 CS	chip select
GPIO9 (opt)	MAX31856 DRDY	data ready
GPIO8 (opt)	MAX31856 FLT	fault
GPIO4	200 ohm -> IRLZ44N gate	LEDC PWM

Avoid strapping pins 0, 3, 45, 46.

Thermocouple

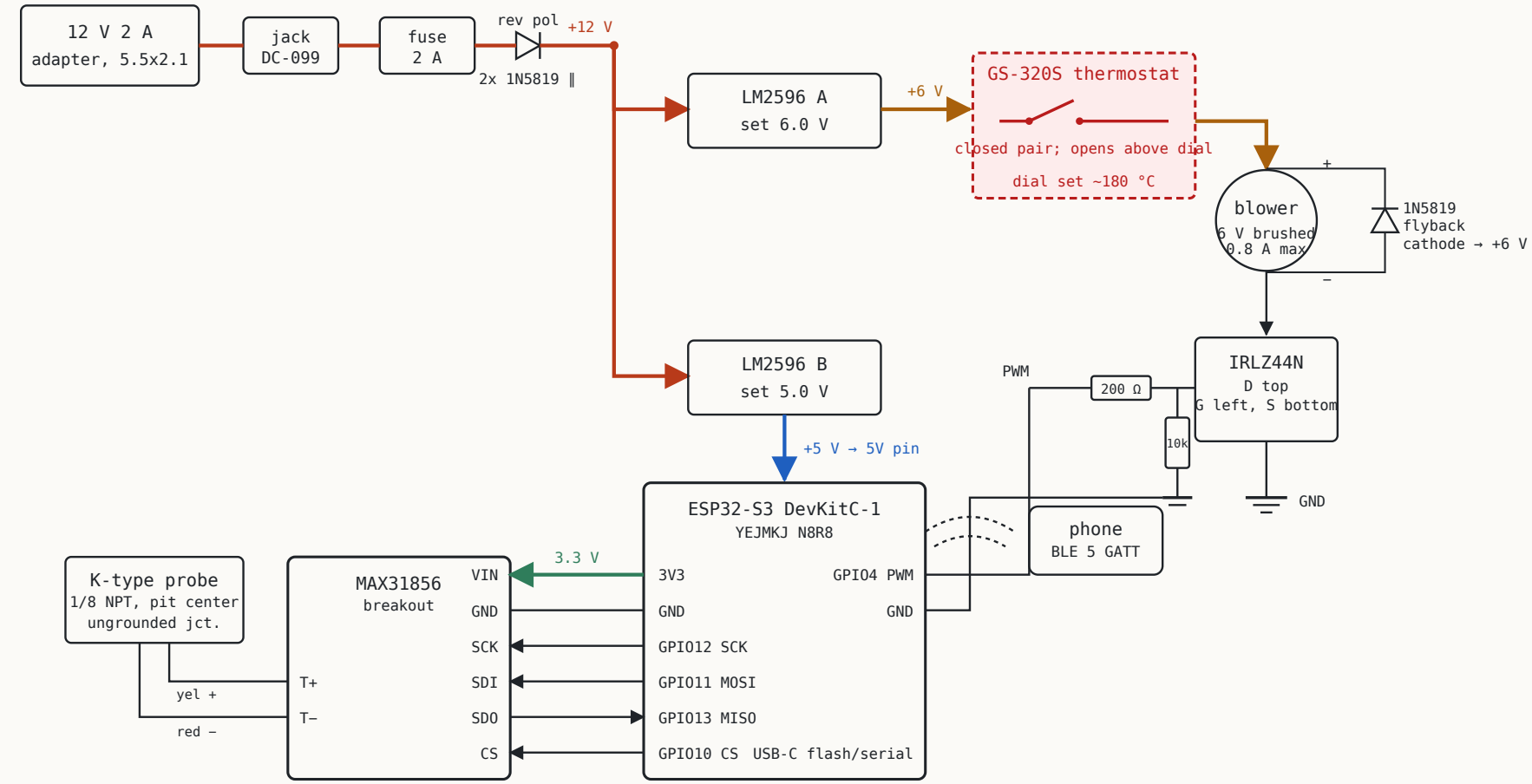
K-type probe -> MAX31856 T+ (yellow) / T- (red). Ungrounded junction required; leads not grounded anywhere; route away from blower rail.

Before power-up

- Set both LM2596 outputs with a meter before connecting loads.
- Identify thermostat closed pair by continuity at room temperature; verify it opens with heat on the bulb.
- Do not feed ESP32 from USB-C and the 5V pin simultaneously unless the board's diode-OR is confirmed.

Smoker Controller Wiring

Phase 1 bench circuit. Pin map and wiring order are on the preceding page.



Common ground: adapter −, both LM2596 GND, ESP32 GND, MAX31856 GND, MOSFET source, 10k pull-down. Star at the buck outputs.
Thermocouple leads are NOT grounded anywhere; route them away from the blower rail.

Power on the top row flows left to right; the thermostat contact sits in the +6 V rail before the blower and opens on over-temperature regardless of firmware. The ESP32 only controls the blower's low side through the MOSFET.

+12 V input · +6 V blower rail · +5 V logic · +3.3 V sensor · hardware cutoff · signal / ground